

Measurement

Toward a Scientific Framework for Interaction Research

Foundational research paper

Abstract

Computational interaction research measures events, traces, task outcomes, subjective reports, system states, and inferred constructs such as discoverability, confidence, workload, belief, strategy, and opportunity. These are often treated as if they belonged to a single evidential category. They do not. Measurement theory distinguishes an empirical relational structure from its numerical representation; psychometrics distinguishes observed responses from latent constructs and requires validity evidence for score interpretation; metrology requires an explicit measurand and uncertainty statement; experimental psychology distinguishes manipulation, observation, and inference; HCI evaluation makes performance context-dependent; and systems engineering emphasizes traceability from requirements to evidence.

This paper concludes that existing measurement theories are sufficient in principle, but computational interaction research requires a shared measurement architecture that integrates them. The proposed architecture defines a measurement claim as a contract among a declared construct, unit and level of analysis, observation model, instrument, coding or transformation rule, numerical scale, uncertainty model, validity argument, and admissible interpretation. It separates direct observations, derived variables, model-based estimates, and latent inferences; forbids interpreting logging precision as construct validity; and requires comparison claims to declare invariance, context, population, task, projection, and tolerance conditions. The paper supplies a measurement ontology, canonical metrics, evidence taxonomy, experimental framework, construct cards, uncertainty budgets, comparability tests, and falsification criteria.

1. Problem Statement

Interaction research routinely reports time, success, error, clicks, paths, ratings, physiological signals, and model predictions. The apparent precision of these quantities can obscure a prior question: what exactly has been measured? A timestamp is directly recorded; hesitation is inferred. A control was displayed; discoverability was not thereby measured. An action was executed; intention, understanding, confidence, or strategy were not directly observed. A task was completed; equal access, equal opportunity, and equivalent interaction were not established.

The central problem is therefore not a shortage of metrics. It is the absence of a common framework that connects theoretical constructs to observations without collapsing observation, transformation, inference, and interpretation. Computational interaction research requires a measurement theory that is implementation-independent, explicit about uncertainty, and compatible with system variables, behavioral traces, subjective reports, and latent constructs.

2. Literature Review Method

The review used a concept-centered synthesis across measurement theory, psychometrics, metrology, statistics, experimental psychology, HCI evaluation, human factors, information theory, and systems engineering. Foundational and primary sources were prioritized: representational measurement theory for numerical representation and scale admissibility; Cronbach and Meehl for construct validity; Messick for validity as an argument about score meaning and use; the Guide to the Expression of Uncertainty in Measurement and NIST guidance for measurands and uncertainty; ISO 9241-11 for context-dependent usability; classical reliability and generalizability theory; signal detection theory; causal inference; and established HCI measures.

The synthesis was constrained by four tests. First, no new construct was retained when an established measurement concept already sufficed. Second, every proposed measurement category had to support operationalization and falsification. Third, direct observations were separated from derived and latent quantities. Fourth, comparison claims were required to state the transformations and scope under which scores remain interpretable.

3. What Measurement Is

Measurement is not merely assigning a number. In representational measurement theory, numerical assignments are justified when the numerical structure preserves relevant relations in an empirical structure, and the permissible transformations of the scale determine which comparisons and statistics remain meaningful. This yields two central requirements: representation and uniqueness. A measure must preserve declared empirical relations, and its interpretation must respect the scale transformations under which that representation remains valid.

Metrology adds the measurand: the quantity intended to be measured. A result is incomplete without specification of the measurand, method, conditions, and uncertainty. Psychometrics adds a further constraint: many interaction constructs are not directly observable. Scores are evidence-bearing observations whose interpretation as measures of a construct requires a nomological and empirical argument.

For computational interaction, measurement is therefore defined as:

A rule-governed process that maps observations of a declared interaction phenomenon into a representational structure, together with an uncertainty statement and a validity argument specifying which interpretations and comparisons are warranted.

This definition distinguishes measurement from logging, counting, classification, estimation, and explanation. Logs may supply observations. Counts may summarize observations. Classifiers may derive labels. Models may estimate latent variables. None of these alone establishes that the resulting value measures the intended construct.

4. Measurement Ontology

The minimum ontology contains the following entities and relations.

Construct: a theoretical property, relation, process, or outcome to be investigated.

Measurand: the specific quantity or attribute intended to be measured in a declared case.

Unit of analysis: the entity or aggregate to which the measurement is attributed, such as event, action, episode, participant, dyad, system, task, session, or population.

Level of analysis: the temporal, organizational, representational, or social level at which the claim is made.

Observation: a recorded state, event, signal, report, or trace element obtained through a declared observation process.

Instrument: the mechanism that produces or records observations, including logger, sensor, questionnaire, protocol, annotator, or system model.

Operationalization: the rule connecting a construct to observable indicators and transformations.

Indicator: an observable or derived variable used as evidence about a construct.

Score: a numerical or categorical representation produced by the measurement procedure.

Scale: the mathematical structure and permissible transformations governing the score.

Estimator: a procedure that infers an unknown quantity from observations.

Uncertainty: quantified incompleteness of knowledge about the measurand or score interpretation.

Validity argument: the evidence and reasoning supporting an intended interpretation and use.

Reliability: consistency of scores under declared replications, not truth or validity.

Comparison scope: the population, task, context, projection, and tolerance under which scores may be compared.

Decision rule: a threshold or mapping from measurements to classifications or actions.

Evidence record: a traceable package containing observations, transformations, assumptions, and uncertainty.

The core relation is not construct -> metric. It is:

Construct -> operationalization -> observation process -> observations -> transformation or estimator -> score -> interpretation.

Validity and uncertainty apply across this chain, not only to the final score.

5. Observable, Derived, and Latent Variables

Interaction measurements should be classified by epistemic status.

Directly recorded variables are produced by a declared instrument with minimal semantic transformation: timestamps, event identifiers, system states, command invocations, pointer coordinates, sensor values, text responses, and explicit ratings. They remain instrument- and boundary-dependent.

Coded observations apply a rule to raw records: error type, action category, recovery event, strategy segment, or successful transition. Coding introduces a model and may require inter-rater or algorithmic reliability.

Derived variables are deterministic or explicitly statistical transformations of observations: duration, path length, transition count, error rate, entropy, precision, recall, edit distance, or proportion successful. Derivation does not by itself validate the construct interpretation.

Model-based estimates are quantities inferred under a probabilistic or computational model: latent state, posterior belief, workload estimate, policy parameter, hazard, random effect, or predicted success probability. Their meaning depends on model assumptions, identifiability, calibration, and fit.

Latent constructs are theoretical variables not directly observed: confidence, understanding, intention, trust, discoverability, cognitive workload, strategy, perceived opportunity, or expertise. They require multiple indicators, experimental interventions, convergent and discriminant evidence, or a validated measurement model. A single behavioral event cannot ordinarily identify a latent construct.

Interpretive conclusions are claims about mechanisms or causes. They are not measurements, even when based on measured variables.

6. Construct Validity

Construct validity is the central problem for interaction measurement because many target constructs are indirect. Cronbach and Meehl require a network of theoretical relations connecting the construct to observations and other constructs. Messick further treats validity as an integrated argument about score meaning and use, rather than a property possessed by an instrument in isolation.

A defensible interaction measurement should address:

Content relevance: indicators sample the intended construct rather than a convenient proxy.

Structural validity: the scoring or latent-variable structure matches the proposed construct structure.

Convergent evidence: different methods intended to measure the same construct show the expected relationship.

Discriminant evidence: the measure is separable from adjacent constructs.

Criterion relations: scores relate to external outcomes as theory predicts.

Known-groups or intervention evidence: manipulations expected to change the construct change the score, while controls do not.

Process evidence: the observation-generating process matches the proposed interpretation.

Generalization: score meaning remains stable across relevant populations, tasks, contexts, and projections.

Consequential validity: intended uses and foreseeable misuses are evaluated where scores guide decisions.

No single correlation establishes validity. Validation is cumulative and claim-specific. A measure may be valid for comparing discovery time within one task yet invalid for ranking general interface quality.

7. Reliability, Reproducibility, and Generalizability

Reliability concerns consistency under repeated measurement conditions. It may include temporal stability, internal consistency, inter-rater agreement, inter-instrument agreement, or repeated-trial consistency. High reliability is necessary for many comparisons but is not sufficient for validity; a logger can record the wrong event with perfect consistency.

Reproducibility requires sufficient specification for independent researchers to implement the measurement process and obtain results consistent within declared uncertainty. Computational interaction studies should preserve event schemas, clocks, task definitions, exclusions, preprocessing, coding manuals, estimator specifications, software versions, and analysis decisions.

Generalizability concerns variation across persons, tasks, items, systems, contexts, and occasions. A score is not automatically an attribute of a person or interface. Many HCI measures are relational properties of users, goals, systems, and contexts. Generalizability theory and multilevel models are therefore more appropriate than treating all variance outside the focal factor as undifferentiated error.

A measurement report should state the intended universe of generalization and identify which facets were sampled, fixed, randomized, or omitted.

8. Uncertainty

Uncertainty in interaction research has several sources.

Instrument uncertainty includes clock resolution, event loss, sensor error, sampling jitter, synchronization drift, and annotation interface constraints.

Operational uncertainty arises from ambiguous event boundaries, task completion rules, segmentation, and competing operational definitions.

Sampling uncertainty concerns persons, tasks, items, sessions, and contexts.

Model uncertainty concerns functional form, priors, unobserved confounding, missing data, identifiability, and estimator choice.

Construct uncertainty concerns whether the indicators support the intended latent interpretation.

Context uncertainty concerns whether measurements transfer across projections, devices, languages, cultures, expertise levels, and environments.

Decision uncertainty concerns thresholds and the consequences of false classifications.

An uncertainty budget should list each source, its treatment, and whether it is quantified, bounded, sensitivity-tested, or unresolved. Confidence intervals alone do not cover model, construct, or scope uncertainty.

9. Canonical Metric Families

No universal scalar interaction metric is supported. The literature instead supports typed metric families.

Temporal metrics: latency, duration, inter-event interval, dwell, recovery time, learning rate, and time-to-discovery.

Outcome metrics: task success, transition success, goal attainment, partial completion, quality, safety, and persistence of effects.

Error metrics: error count, error probability, severity, detectability, correction rate, escalation, and residual harm.

Path and sequence metrics: action count, path length, edit distance, state coverage, branching, backtracking, sequence entropy, and conformance to a declared model.

Opportunity metrics: precision and recall of believed actions against executable actions, discovery probability, exposure coverage, signaling calibration, reachable-outcome coverage, and recovery visibility.

Information metrics: entropy, mutual information, uncertainty reduction, channel capacity, and coding cost, only when the random variables and probabilistic assumptions are defined. Shannon information does not measure meaning, understanding, or usefulness by itself.

Resource metrics: physical effort, attentional demand, memory demand, computational cost, energy, bandwidth, and coordination load.

Subjective measures: perceived workload, confidence, trust, satisfaction, risk, usability, and perceived reversibility. These are reports under a measurement model, not transparent readings of internal state.

Comparative metrics: effect size, equivalence bounds, calibration difference, distributional distance, and invariance tests.

Each metric must be tied to a construct card. Metrics should not be combined into an index unless the aggregation rule and trade-offs are theoretically justified.

10. Measurement Contracts

Every reported interaction measure should be accompanied by a measurement contract containing:

1. Construct and intended interpretation.
2. Measurand and unit of analysis.
3. Scope conditions: population, task, system, projection, context, and time.
4. Observation model and instrument.
5. Event, state, and boundary definitions.
6. Operationalization and indicators.
7. Transformation, coding, or estimator.
8. Scale type and permissible operations.
9. Missingness, exclusions, and preprocessing.
10. Reliability evidence.
11. Validity evidence.
12. Uncertainty budget.
13. Comparison and invariance assumptions.
14. Failure conditions and prohibited interpretations.
15. Reproducibility artifacts.

This contract is the minimum reusable unit of measurement specification. It makes the evidential chain inspectable and supports later interaction-similarity and comparative-interaction research.

11. Comparability and Measurement Invariance

Two scores are comparable only when they preserve an intended interpretation across the cases being compared. Numerical equality is insufficient. The same completion time can arise from different task boundaries, clocks, path constraints, or strategies. The same questionnaire total can have different factor structures or response meanings across populations.

Comparability requires at least:

Construct identity: the same construct interpretation is intended.

Operational correspondence: indicators and coding rules have equivalent roles.

Scale compatibility: permissible transformations and units are aligned.

Measurement invariance: item or indicator relations remain stable where latent scores are compared.

Context declaration: task, goals, authority, projection, and environment are controlled or modeled.

Sampling alignment: populations and units are comparable or adjusted.

Uncertainty compatibility: uncertainty is reported on commensurate terms.

Tolerance: equivalence is defined by a prespecified practically meaningful bound.

When these conditions fail, the correct result is typed non-comparability, not forced normalization.

12. Experimental Framework

The recommended experimental framework separates construct, intervention, measurement, and inference.

Phase 1 - Theory and construct specification. Define the construct, competing explanations, unit and level of analysis, and causal role.

Phase 2 - Measurement design. Create construct cards, select indicators, define events and boundaries, specify instruments, and preregister uncertainty and exclusion rules.

Phase 3 - Manipulation and comparison design. Specify treatment, control, manipulation checks, nuisance variables, and minimum meaningful effects or equivalence bounds.

Phase 4 - Data generation. Record raw observations with provenance, synchronized clocks, immutable event identifiers, and declared missingness.

Phase 5 - Transformation and coding. Apply versioned rules; preserve raw-to-derived traceability; assess rater or algorithmic agreement.

Phase 6 - Estimation. Fit models appropriate to nesting, repeated measures, censoring, measurement error, and uncertainty. Avoid treating participant or task variance as independent observations.

Phase 7 - Validation. Test convergent, discriminant, structural, intervention, criterion, and generalization evidence.

Phase 8 - Sensitivity and falsification. Test alternative operationalizations, priors, exclusions, model forms, and rival mechanisms.

Phase 9 - Reporting. Publish the measurement contract, uncertainty budget, analysis specification, and prohibited interpretations.

13. Evidence Taxonomy

Evidence should be reported by level.

Level 0 - Raw record: minimally processed event, signal, state, or report.

Level 1 - Verified observation: record checked for provenance, integrity, timing, and instrument behavior.

Level 2 - Coded observation: category assigned under a documented rule.

Level 3 - Derived measure: deterministic or statistical summary of observations.

Level 4 - Model estimate: parameter, posterior, or latent score inferred under a declared model.

Level 5 - Construct evidence: multiple measures support an interpretation of a theoretical construct.

Level 6 - Mechanism evidence: interventions, temporal ordering, and rival tests support a generating process.

Level 7 - Generalized claim: evidence supports transfer across declared populations, tasks, systems, or projections.

Higher levels depend on lower levels but do not become direct observations. A mechanism claim should remain labeled as inference even when strongly supported.

14. Falsification Tests

The framework should be narrowed or rejected if the following occur.

Measurement redundancy: established measurement theory can already express every requirement without any interaction-specific coordination artifact. In that case only a domain checklist, not a new framework, is warranted.

Non-operational constructs: central interaction constructs cannot be connected to observable indicators with discriminant validity.

No incremental validity: specialized opportunity, projection, or interaction metrics fail to explain or predict outcomes beyond ordinary task, system, and participant variables.

Invariance failure: measures change interpretation across projections, populations, or tasks despite apparent numerical comparability.

Observer dependence without calibration: event boundaries and coding cannot achieve reproducible agreement.

Uncertainty dominance: measurement and model uncertainty exceed the effects under investigation.

Causal ambiguity: proposed mechanisms remain indistinguishable from simpler rival explanations under intervention.

Metric instability: conclusions reverse under equally defensible operationalizations or preprocessing choices.

The framework survives only if it improves traceability, construct validity, and comparative inference relative to untyped metric reporting.

15. Threats to Validity

The synthesis risks over-formalizing phenomena that are situated, socially negotiated, or emergent. Not every meaningful interaction property is naturally scalar. Qualitative observations may provide stronger construct evidence than forced quantification.

Measurement contracts can create procedural compliance without genuine validity. Explicit fields do not substitute for theory or empirical validation.

The framework inherits disputes within measurement theory, including whether all measurement requires numerical structural representation and how causal accounts of measurement relate to representational accounts.

HCI constructs often depend on context of use. Generalization across tasks, cultures, languages, accessibility conditions, and expertise may be limited. Measurement invariance may fail for theoretically meaningful reasons rather than instrument defects.

Instrumentation can change the phenomenon being measured through reactivity, latency, logging overhead, surveillance effects, or task restructuring.

Finally, evidence quality is not reducible to a linear hierarchy. Different questions require different forms of evidence; the taxonomy records dependency and inferential distance, not universal superiority.

16. Implications for Interaction Research

The measurement framework completes the transition from conceptual ontology to empirical research. Prior work distinguishes ontology, semantics, representation, interpretation, opportunity, interaction, and behavior. Measurement adds the rules by which claims about these constructs can be connected to evidence without collapsing them.

For representation research, it requires typed equivalence tests and invariance conditions. For opportunity research, it separates executable, exposed, signaled, believed, selected, and successful actions into different measurands. For behavior research, it converts traces into evidence through declared segmentation and coding. For interaction comparison, it supplies equivalence bounds and comparability tests. For prediction, it separates measurement quality from model performance.

The central implication is that computational interaction research should stop treating metric labels as self-interpreting. Time, error, success, workload, trust, discoverability, and usability are valid only through explicit measurement contracts.

17. Conclusion

Existing measurement theory, psychometrics, metrology, statistics, experimental psychology, HCI evaluation, and systems engineering are sufficient to explain how interaction phenomena can be measured. No new foundational theory of measurement is required.

A domain-specific synthesis is nevertheless necessary because interaction research combines system events, behavioral traces, subjective reports, relational constructs, and latent interpretations in ways that invite category errors. The warranted contribution is a shared measurement architecture, not a new measurement primitive.

The primary conclusion is:

Interaction phenomena should be measured through explicit contracts linking constructs to observation processes, transformations, uncertainty, validity evidence, and comparison scope. Observations, derived metrics, latent estimates, and explanatory conclusions must remain typed and traceable throughout the evidential chain.

18. Reusable Artifacts

A. Measurement Ontology

Entity	Definition	Role
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Toward a Scientific Framework for Interaction Research

Construct	The theoretical property, relation, process, or outcome under investigation.	Theory
Measurand	The specific quantity intended to be measured in a declared case.	Specification
Observation	Recorded event, state, signal, or report.	Empirical
Instrument	Mechanism producing or recording observations.	Method
Indicator	Observable or derived evidence linked to a construct.	Operationalization
Score	Representational output of a measurement procedure.	Representation
Estimator	Rule for inferring an unknown quantity from observations.	Model
Uncertainty	Incomplete knowledge about measurand or interpretation.	Epistemic
Validity argument	Evidence supporting an intended score interpretation and use.	Inference
Comparison scope	Conditions under which values may be compared.	Boundary

B. Canonical Metric Registry

Construct family	Candidate metrics	Required controls	Prohibited inference
Discovery	Unaided discovery probability; time-to-discovery; prompt level	Exposure/signaling held or modeled; prior knowledge measured	Visibility alone is not discovery
Believed action set	Precision/recall against executable action set; confidence calibration	Judgment elicited before feedback; executable set independently verified	Attempts alone do not identify belief
Behavioral path	Transition count; path cost; entropy; backtracking; conformance	Common state/action ontology and segmentation	Shorter is not universally better
Recovery	Detection time; repair initiation; successful recovery; residual effect	Error severity and actual recovery graph declared	Undo availability is not perceived reversibility
Performance	Effectiveness, efficiency, quality, safety	Users, goals, context, and success criteria declared	Task completion is not general usability
Subjective state	Rating scale or latent score with reliability and validity evidence	Response process and invariance tested	Report is evidence, not transparent internal state
Representation comparison	Equivalence bounds; effect sizes; distributional distance	Semantic target and task held constant	Informational equivalence does not imply behavioral equivalence

C. Evidence Taxonomy

Level	Meaning	Example	Minimum requirement
0 Raw record	Event, signal, state, report	Logger packet; audio sample	Provenance and integrity
1 Verified observation	Validated record	Synchronized event timestamp	Instrument behavior checked
2 Coded observation	Rule-applied category	Error type; action class	Coding reliability
3 Derived measure	Deterministic/statistical summary	Duration; success rate	Derivation traceable
4 Model estimate	Quantity inferred under model	Latent workload; hazard	Fit, identifiability, calibration
5 Construct evidence	Multiple indicators support construct	Discoverability evidence	Convergent/discriminant validity
6 Mechanism evidence	Intervention supports process	Signaling changes discovery via belief	Rival explanations tested
7 Generalized claim	Claim extends across scope	Cross-projection invariance	Sampling and invariance evidence

D. Construct Card

- Construct name and definition
- Adjacent constructs and discriminant boundary
- Causal or explanatory role
- Unit and level of analysis
- Candidate indicators
- Expected convergent and discriminant relations
- Manipulations expected to change the construct
- Conditions expected not to change it

- Known confounds
- Permitted interpretations
- Prohibited interpretations
- Falsification conditions

E. Measurement Contract

1. Construct and intended interpretation
2. Measurand
3. Unit and level of analysis
4. Population, task, system, projection, context, and time scope
5. Observation model and instrument
6. Event, state, and boundary definitions
7. Operationalization and indicators
8. Transformation, coding, or estimator
9. Scale type and permissible operations
10. Missingness, exclusions, and preprocessing
11. Reliability evidence
12. Validity evidence
13. Uncertainty budget
14. Comparison and invariance assumptions
15. Failure conditions and prohibited interpretations
16. Reproducibility artifacts

F. Uncertainty Budget Template

Source	Mechanism	Direction/magnitude	Treatment	Residual risk
Instrument				
Clock/synchronization				
Boundary/segmentation				
Coding				
Sampling				
Missingness				
Model form				
Construct interpretation				
Context transfer				
Decision threshold				

G. Experimental Protocol

1. Define construct, rival explanations, unit, and level.
2. Create construct card and measurement contract.
3. Specify treatment, control, manipulation check, nuisance variables, and equivalence bounds.
4. Capture raw observations with provenance and synchronized timing.
5. Apply versioned coding and transformation rules.
6. Estimate with models appropriate to nesting, censoring, and measurement error.
7. Validate score interpretation using multiple evidence sources.
8. Run sensitivity, invariance, and falsification analyses.
9. Report uncertainty, scope, prohibited interpretations, and reproducibility artifacts.

19. References

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